External Validation of an Interpretable Integer Scorecard versus Logistic Regression and Tree Ensembles for Ovarian Cancer Risk Prediction

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***Abstract*—Preoperative blood tests decide whether an ovarian mass needs cancer surgery; a model that degrades silently at a new hospital can delay surgery or trigger unnecessary operations. Eight models—three published rules (a CA125 cutoff, ROMA, CPH-I) and five fitted models (a logistic regression, an integer scorecard, three tree ensembles)—were trained or fixed on a Chinese cohort (349 patients) and applied, unchanged, to two other Asian cohorts (West China, 380; Japan, 177). On the 100 West China patients with measured biomarkers (63% cancer prevalence), the scorecard (AUROC 0.971) was numerically highest, significantly exceeding CatBoost (0.910) and Random Forest (0.930) after multiple-comparison correction but not the logistic regression (0.941), ROMA (0.965), or XGBoost (0.946). The logistic regression tied or beat the scorecard on the imputed cohort (0.936 vs. 0.929), under multiple imputation (0.926 vs. 0.853), and under measurement noise; on the imputed cohort the scorecard missed 47 of 188 cancers at its fixed high-risk tier versus 82 (ROMA, an imputation artifact) and 48 (CPH-I) at their published cutpoints. A simulation (illustrative counterexample only) showed the flexible models' advantage diminishing and reversing under simulated drift. Within these small cohorts the evidence therefore supports parity: the scorecard matched the linear models and outperformed two of the three tuned ensembles on the complete cases—an outcome-selected subset skewed toward high-marker cancers, so that margin may reflect case mix, not algorithmic superiority—and its distinctive asset is hand-computability at the bedside, not predictive superiority; generalization requires larger validation studies.**

***Keywords*—ovarian cancer; risk prediction; scorecard; external validation; transportability**

I. Introduction

Ovarian cancer is the most lethal gynecologic cancer, and most cases are found late. When a pelvic mass is found, referral to a specialist cancer surgeon depends on preoperative risk, and errors in either direction are costly. That decision uses two cheaply measured blood proteins, CA125 (cancer antigen 125) and HE4 (human epididymis protein 4), combined by fixed formulas such as the Risk of Ovarian Malignancy Algorithm (ROMA) [1] and the Copenhagen Index (CPH-I) [2], or by a fitted model. Machine learning models have been proposed for the same task, usually with strong results on training data but little external testing; [3] found no performance benefit over logistic regression across clinical prediction models.

Lu et al. [4] trained machine-learning models on the Changzhou cohort reused here, selecting HE4 and carcinoembryonic antigen (CEA) for a decision tree that beat ROMA and logistic regression on a held-out split. Our work adds a hand-computable scorecard, applies every model frozen to two Asian cohorts, and asks: does model complexity produce performance that survives a move to a new hospital, and can a bedside-computable scorecard match the logistic regression and the tree ensembles? Interpretability arguments favor such simple models in high-stakes medicine [5]; we measured discrimination, calibration, and decision consequences, plus a simulation of when complex models help.

II. Methods

A. Data

Three public, anonymized datasets were used [6]–[8] (IRB approval was not required for re-analysis of public, de-identified data). The cohorts are a training cohort from Changzhou, Jiangsu, China (349 patients; 171 ovarian cancers, 178 benign masses; Mendeley Data [6], doi:10.17632/th7fztbrv9.11, from Lu et al. [4]), a second Chinese cohort from the West China Second University Hospital, Chengdu (380 patients; 188 cancers, 192 benign cysts; figshare [7], doi:10.6084/m9.figshare.28831256), and a Japanese cohort (177 patients; 27 cancers, 150 healthy women; Karger figshare [8], doi:10.6084/m9.figshare.24235450). Each patient has four features: CA125 (U/mL), HE4 (pmol/L), age, and menopausal status (recorded in all three cohorts); `src/clean_data.py` harmonizes the source files into the processed files used by every analysis. CEA, Lu et al.'s key feature [4], is absent from the two external cohorts, so fitted models use the four common features. Scattered training missing values (17 CA125, 20 HE4) were filled by class-stratified k-nearest-neighbor imputation ($k = \min(5, \text{class size})$, applied once before modeling). Externally, HE4 was not measured for 74% of West China patients and age was not recorded for 85% of Japan patients; these external gaps were filled with the frozen training-set medians (HE4 54.5 pmol/L, CA125 49.3 U/mL, age 45.0 years), so no validation statistic is used for imputation. Because the Japanese cohort contains 150 healthy women and only 27 cancers, and

performance depends on the base rate among patients tested [9], it is a specificity stress test, not external validation; its AUROC is never pooled with West China's. The primary West China analyses use the 100 patients with measured HE4 (63 cancers, 37 benign); the datasets do not record why HE4 was measured only there. Relative to the complete cases, the 280 excluded patients (125 cancers, 155 benign) had similar age (42.2 vs. 45.2 years) and menopausal rates (31.8% vs. 31.0%), lower CA125 (median 60.7 vs. 95.1 U/mL), and lower cancer prevalence: 44.6% versus 63.0% (Fisher exact test $p = 0.002$). The complete-case benchmark is therefore selected on the outcome, and all of its estimates are complete-case estimates at 63% prevalence. The 74%-imputed cohort ($N = 380$) is a sensitivity analysis for the fitted models, but for ROMA it is not a valid evaluation: with HE4 fixed at one training median, premenopausal ROMA moves only $0.0626 \times \ln(5000/5) = 0.43$ logits across a 1,000-fold CA125 change, and none of the 191 imputed premenopausal patients crosses the 13.1% cutoff at any observed CA125 value—ROMA's imputed-cohort AUROC and operating points are imputation artifacts, reported only for completeness. No-HE4 refits and MICE AUROCs bound this effect (Section III-C).

B. Models

The five fitted models used the same four features; the three published rules use their published inputs (ROMA: HE4, CA125, menopause; CPH-I: HE4, CA125, age; CA125 rule: CA125 alone). (1) *Integer scorecard*: each continuous feature was rank-transformed and cut into training-set tertiles (frozen cutpoints: CA125 24.3/165.0 U/mL, HE4 45.8/92.7 pmol/L, age 37.0/52.0 years), then one-hot encoded and fitted with an L2-regularized logistic regression ($C = 1.5$); each coefficient β_i was then rescaled to the nearest whole number $w_i = \text{round}(5 \times \beta_i / \max_j |\beta_j|) \in \{-5, \ldots, +5\}$, with $|\beta_i| < 0.01$ set to zero. The score is the sum of these feature points (range −7 to +9: CA125 −1/0/+2, HE4 −4/−1/+5, age −2/0/+2, menopause 0, a single bin because the feature is binary), can be computed by hand, and contains seven fitted integer parameters (two, three, and two nonzero bin weights for CA125, HE4, and age, respectively; the intercept rounds to zero and is unused). HE4 spans −4 to +5 versus −1 to +2 for CA125, matching HE4's cancer specificity and CA125's elevation in endometriosis; the menopausal coefficient rounded to zero; the fixed high-risk tier is score ≥ +1, and a training-fitted logistic map converts scores to probabilities, both frozen before external evaluation. (2) *Logistic regression* on standardized features (no log transform; CA125 and HE4 standardized on the raw scale). (3–4) the published formulas *ROMA* and *CPH-I*, computed on raw values: premenopausal ROMA is $-12.0 + 2.38 \ln(\text{HE4}) + 0.0626 \ln(\text{CA125})$ (postmenopausal coefficients −8.09, 1.04, 0.732), logistic-transformed, with published cutoffs 13.1% (pre) and 27.7% (post) [1]; CPH-I is $-14.0647 + 1.0649 \log_2(\text{HE4}) + 0.6050 \log_2(\text{CA125}) + 0.2672 \times \text{age}/10$, likewise logistic-transformed, with cutoff 7% [2]. Units: HE4 pmol/L, CA125 U/mL, age years; within-cohort recalibration means fitting a logistic regression of the outcome on the model's output (score or probability) as the sole covariate—a monotone rescaling—on West China. (5) a single *CA125 ≥ 35 U/mL* rule. (6–8) *XGBoost* [10], *CatBoost* [11], and *Random Forest* [12], evaluated under two settings: Setting A keeps each package's default hyperparameters, and Setting B tunes each ensemble by nested cross-validation, with all tuning decisions confined to training folds (XGBoost 100 trees, depth 3; CatBoost 500 iterations, depth 4; Random Forest 500 trees, depth 6). The primary comparison uses Setting B (Table I, Fig. 1); Setting A values are in its caption; the five fitted models were trained once on the Chinese cohort and frozen before any external evaluation (the published rules are fixed by definition).

C. Evaluation

Discrimination was measured by the area under the ROC curve (AUROC), with 95% confidence intervals and paired bootstrap tests between models (2,000 patient-level resamples). The prespecified primary comparison is the scorecard versus ROMA on the 100 complete cases; the same comparisons on the 74%-imputed cohort are sensitivity analyses. All other comparisons are secondary, and effect sizes with 95% CIs are emphasized over p-values, with Holm–Bonferroni adjustment across the six comparisons of Section III-A. Calibration was measured with the Brier score, calibration-in-the-large (CITL; the logit of the mean predicted risk minus the logit of the observed prevalence, negative when under-predicting), the calibration slope (the logistic coefficient of the outcome on the logit of predicted risk; 1 = perfect), and the expected calibration error (ECE), the mean absolute difference between observed and predicted risk across ten equal-width probability bins [13]; frozen calibration on the 100 complete cases is reported in Section III-C. Decision value used a weighted-harm analysis (Section III-B) [14]: thresholds were swept over 1–60% (Fig. 2 sweeps 1–99%), and for exchange rates $w \in \{1, 5, 10, 50\}$ the threshold minimizing $H = w \times \text{missed} + \text{unnecessary}$ was identified; for a perfectly calibrated model this minimizer is $t = 1/(1 + w)$ (50% down to 2.0%), probing from indifference ($w = 1$) toward weighting missed cancers heavily ($w = 50$). Q1 asks whether each model transports unchanged; Q2 how well it performs after local recalibration. The weights are hypothetical scenarios, not validated clinical utilities; all design choices (features, primary comparison, weights) preceded external evaluation, and only the scorecard's bin count and regularization were selected among 12 candidates by repeated CV (Section III-E). The primary harm comparison applies within-cohort recalibration to all five models, so none gains from refitting on the validation outcome; published-cutpoint operating points are reported separately. The retraining experiment used stratified training subsets ($n = 60$ to 349, 10 draws), evaluated unchanged on the imputed West China cohort. Robustness was tested with 10% and 30% multiplicative, additive, and combined noise, 5% gross-error contamination, and ±10% bin-boundary shifts; the noise grid, retraining, and no-HE4 refits use the ensembles at package defaults (Setting A). A MICE-style check imputed West China's features by chained-equation regression (10 iterations, outcome excluded, one dataset). Analyses used Python 3.13 with scikit-learn 1.7.2, XGBoost 3.3.0, and CatBoost 1.2.10 (seed 42, stratified

five-fold cross-validation, 20 simulation replicates); Setting A (defaults) is XGBoost 100 trees depth 6, CatBoost 1,000 iterations depth 6, Random Forest 100 trees uncapped. Following calls for released, reproducible pipelines [15], our pipeline (`reproduce.sh`, pinned dependencies) is available at https://github.com/angelayaoo/Ovarian-Cancer-RiskPrediction.

D. Simulation

The simulation is an illustrative counterexample, not an explanation of the observed real-world results: it varies drift (d) independently of covariate shift (s) and noise. A simulation calibrated to the Chinese data drew labels from a mixture distribution and biomarkers from the corresponding class-conditional lognormal distributions as a proposal, then generated the outcome from the logistic model

$$\begin{aligned}\eta(z_1, z_2, z_3) = &-0.65 + 1.8z_1 + 1.0z_2 + 0.5z_3 \\ &+ (1-d)\big(1.6z_1z_2 + 1.2\sin(1.7z_1)\big),\end{aligned} \tag{1}$$

where z_1, z_2, and z_3 are the standardized log CA125, log HE4, and age, respectively, and $d \in [0, 1]$ sets the drift level: $d = 0$ keeps the interaction and nonlinear terms, $d = 1$ removes them on the test side, and intermediate values scale them by $1 - d$, as if the biomarker–cancer relationship partially changed between hospitals. The grid crossed training sizes $n \in \{100, 200, 400, 800\}$, shifts $s \in \{0, 0.5, 1.0\}$, noise $\{0\%, 50\%\}$, and drift levels $d \in \{0, 1\}$, with 20 replicates per cell. Shift s shrinks class-conditional means toward the pooled mean and inflates variances (less separation); noise adds Gaussian error of 50% of each biomarker's standard deviation to CA125 and HE4.

III. Results

A. Discrimination

On the 100 West China patients with measured HE4 (63 cancers), the scorecard was numerically highest, with ROMA, tuned XGBoost, and the logistic regression not significantly different in paired bootstrap tests (Table I, West CC column; Fig. 1). This benchmark is selected on the outcome (63.0% vs. 44.6% cancer prevalence; Fisher $p = 0.002$), and 0.971 exceeds the scorecard's internal CV AUROC of 0.906 (Section III-E); it is a complete-case estimate at 63% prevalence on a cancer-enriched subset (median CA125 95.1 vs. 60.7 U/mL), not evidence of superior external performance. On the 100 complete cases, the primary comparison gave a scorecard-minus-ROMA difference of +0.006 AUROC (95% CI −0.019 to 0.032; $p = 0.65$); the scorecard also did not differ from the logistic regression (+0.030; $p = 0.09$) or XGBoost (+0.024; $p = 0.08$), and exceeded CPH-I (+0.040; $p = 0.03$), CatBoost (+0.058; $p = 0.004$), and Random Forest (+0.040; $p = 0.006$); after Holm–Bonferroni, CatBoost (0.02) and Random Forest (0.03) remained significant. (Paired differences are bootstrap means, not differences of Table I means.) On the imputed cohort (sensitivity analysis), the differences all favored the scorecard (vs. ROMA +0.043—an imputation artifact—CPH-I +0.061, XGBoost +0.040, CatBoost +0.052, Random Forest +0.034; all CIs excluded zero) except the logistic regression (−0.007; $p = 0.48$), which was highest there (0.936 vs. 0.929; ensembles 0.877–0.895).

TABLE I

AUROC (95% CI in parentheses) by model and cohort, bootstrap means over 2,000 resamples. West CC (complete cases): the 100 West China patients with measured HE4 (63 cancers, 63.0% prevalence; primary external benchmark, selected on the outcome); West: the full 74%-imputed cohort ($N = 380$, sensitivity analysis); Japan: healthy volunteers (specificity stress test; CPH-I computed with age imputed for 151/177 patients). ROMA's West column is an imputation artifact (constant HE4), not a valid evaluation of the formula. Setting A (defaults) ensembles, in the order XGBoost, CatBoost, Random Forest, scored 0.921/0.922/0.910 (CC), 0.887/0.879/0.879 (West), 0.736/0.750/0.794 (Japan). Bold: highest per column and values whose paired bootstrap difference from it has $p > 0.05$.

Model	West CC (n=100)	West (N=380)	Japan
Scorecard	**0.971 (0.936–0.995)**	**0.929 (0.902–0.955)**	**0.845 (0.714–0.949)**
Logistic reg.	**0.941 (0.887–0.983)**	**0.936 (0.908–0.961)**	**0.806 (0.666–0.926)**
ROMA	**0.965 (0.930–0.992)**	0.886 (0.848–0.921)	0.799 (0.666–0.908)
CPH-I	0.929 (0.875–0.973)	0.868 (0.828–0.905)	**0.920 (0.852–0.973)**
CA125 rule	0.694 (0.605–0.785)	0.642 (0.597–0.684)	0.781 (0.685–0.874)
XGBoost	**0.946 (0.897–0.984)**	0.890 (0.853–0.925)	**0.840 (0.711–0.944)**
CatBoost	0.910 (0.845–0.962)	0.877 (0.836–0.915)	0.772 (0.650–0.880)
Random Forest	0.930 (0.871–0.979)	0.895 (0.858–0.931)	0.819 (0.692–0.926)

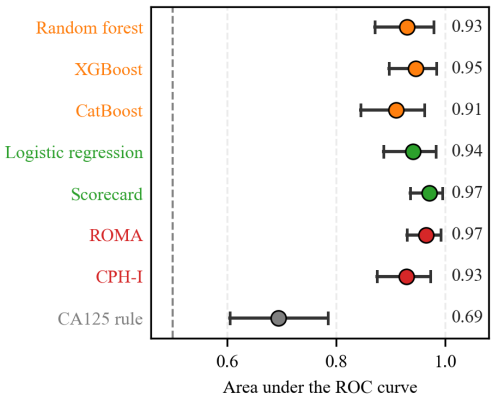


Fig. 1. External validation AUROC, 100 West China patients with measured HE4 (63 cancers; 63% prevalence); whiskers: bootstrap 95% CIs; green = linear, red = formulas, orange = tuned trees, grey = rule.

In the Japan specificity stress test ($n = 177$), CPH-I was highest (0.920; Table I, Japan column; its age is the frozen training median for 151/177 patients). As a specificity stress test, its key operating points are false-positive rates among the 150 healthy volunteers at the fixed cutpoints: 0/150 (scorecard tier, ROMA), 1/150 (0.7%, CPH-I), 10/150 (6.7%, CA125 rule). Within Japan, 85.3% of healthy women but 18.5% of cancer patients fell into both markers' lowest frozen bins; volunteers had far lower markers than West China patients (median CA125 13.9 vs. 65.1 U/mL; HE4 32.5 vs. 54.5 pmol/L, the latter the frozen imputation value).

B. Decision consequences

In the primary weighted-harm analysis (Q2; Table II, Fig. 2), all five models were within-cohort recalibrated, so no model gains from refitting on the validation outcome. The logistic regression's minimum harm was lowest up to $w = 10$ (9.7, 29.5, 39.7); at $w = 50$, recalibrated ROMA was lowest (45.8), then the logistic regression (46.3), CPH-I (50.3), and XGBoost

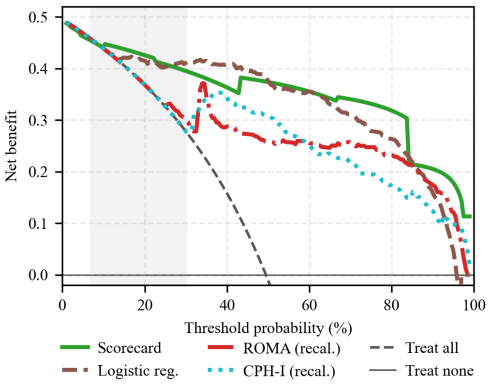


Fig. 2. Decision curve analysis on West China (N = 380, 49.5% cancer prevalence): net benefit per patient versus the decision threshold in percent, with treat-all (dashed; 192/380 benign) and treat-none (the horizontal axis) references; the shaded band is the published-cutpoint range (7–30%). All shown models are within-cohort recalibrated; XGBoost is omitted (minimum harms in Table II).

TABLE II

MINIMUM NET HARM ON WEST CHINA (N = 380, 49.5% CANCER PREVALENCE) AT EXCHANGE-RATE WEIGHT w: HARM = w×MISSED+UNNECESSARY PER 100 WOMEN, MINIMIZED OVER THE SWEPT 1–60% THRESHOLDS (WHICH EXCLUDE TREAT-ALL AT THRESHOLD 0); TREAT-ALL = 192/380 = 50.5 AT EVERY WEIGHT (LOWER IS BETTER). Q2 ROWS: ALL FIVE MODELS WITHIN-COHORT RECALIBRATED (XGBOOST AT PACKAGE DEFAULTS); FROZEN (AS-FITTED) HARMS: LOGISTIC REGRESSION 9.7/29.2/39.7/46.6, XGBOOST 14.7/44.7/66.1/215.5; Q1 PUBLISHED-SCALE HARMS: ROMA 15.0/43.4/45.8/45.8, CPH-I 15.0/51.3/64.7/159.5.

Model	w=1	w=5	w=10	w=50
Logistic reg.	**9.7**	**29.5**	**39.7**	46.3
Scorecard	12.1	30.5	45.0	59.5
XGBoost	14.7	45.8	50.5	50.5
ROMA	14.7	42.9	45.8	**45.8**
CPH-I	15.0	50.3	50.3	50.3
Treat all (reference)	50.5	50.5	50.5	50.5

(50.5); minima are over the swept 1–60% thresholds and can exceed treat-all (threshold 0), as the scorecard does at w = 50 (59.5). Among the three recalibrated rules the scorecard was lowest up to w = 10 (12.1 vs. 14.7/15.0; 45.0 vs. 45.8/50.3) and fell behind at w = 50 (59.5 vs. 45.8/50.3). Recalibration barely moved the logistic regression (frozen 9.7/29.2/39.7/46.6) but transformed XGBoost at w = 10/50 (66.1/215.5 vs. 50.5/50.5), so a frozen-versus-recalibrated comparison would have been unfair to XGBoost. All Table II/Fig. 2 quantities are per 100 women at 49.5% prevalence (treat-all = 192/380 = 50.5) [9]; the formula rows use the imputed cohort, where ROMA's input is partly the imputation constant.

As the Q1 consequence (unchanged transport) on the imputed cohort, the scorecard's high-risk tier (score ≥ +1) missed 47 of 188 cancers (75.0% sensitivity, 97.4% specificity), versus 82 (ROMA; 56.4%, 96.9%; an imputation artifact) and 48 (CPH-I; 74.5%, 94.8%) at the published cutpoints—a one-patient gap to CPH-I. On the 100 measured-HE4 patients, the scorecard and ROMA each missed 13 of 63 cancers (79.4%, 100%), CPH-I 13 (79.4%, 97.3%), and the CA125 rule 8 (87.3%, 51.4%).

C. Calibration and missing-data checks

For calibration on the imputed cohort, the recalibrated scorecard had Brier 0.097, slope 1.00 (guaranteed by the refit—the asymmetry flagged in Section III-B), and ECE 0.042, versus its frozen map (0.116, 2.02, 0.128), the frozen logistic regression (0.111, 1.44, 0.135), ROMA (0.209, 1.53, 0.220), and CPH-I (0.259, 1.11, 0.303), the formulas under-predicting risk (CITL −0.95, −1.42). On the 100 complete cases (all models frozen), Brier/slope/ECE were 0.077/1.77/0.084 (scorecard), 0.105/1.19/0.130 (logistic regression), 0.154/1.77/0.212 (ROMA), and 0.211/0.99/0.272 (CPH-I): the frozen scorecard had the lowest Brier and ECE on measured biology, though its slope was the farthest from 1. Refitting without HE4 gave 0.925 (scorecard), 0.932 (logistic), 0.891 (XGBoost, defaults). The MICE-style imputation (Section II-C) degraded the scorecard's tertile binning and reversed the ensemble ordering, the logistic regression highest throughout; the full MICE AUROC set (Setting B trees) was 0.926 (logistic regression), 0.865 (CatBoost), 0.856 (Random Forest), 0.853 (scorecard), 0.848 (CPH-I), 0.818 (XGBoost), 0.817 (ROMA), and 0.642 (CA125 rule).

D. Robustness and simulation

Under 30% multiplicative noise (West China, N = 380; ensembles at package defaults), AUROC losses were 0.040 (scorecard), 0.002 (logistic regression), 0.074 (ROMA), and 0.086 (XGBoost). Across the full robustness grid of Section II-C (10%/30% multiplicative/additive/combined noise, 5% gross-error contamination), scorecard 0.869–0.921, logistic regression 0.932–0.936, ROMA 0.791–0.871, CPH-I 0.839–0.868, ensembles 0.775–0.882. Shifting the frozen cutpoints by ±10% moved the scorecard's AUROC by at most 0.016 (0.913–0.929); retraining on 60–349 patients (ten draws) gave, at full size, logistic regression 0.936, scorecard 0.929, ensembles 0.878–0.887.

In the simulation (Fig. 3), the ensemble-average minus-scorecard gap averaged +0.015 at d = 0 of (1) and −0.015 at d = 1 across all 24 cells (+0.018/−0.018 in the 50%-noise slice shown, 12 of 24 cells per drift level); at d = 1 the scorecard had the highest AUROC in 22 of 24 cells. That regime favors the scorecard by construction—at d = 1, (1) is additive in the three features, removing the interactions and nonlinearity that flexible models learn (the scorecard's bins are additive too)—so the cells measure loss of the ensembles' learned structure, not scorecard superiority. The XGBoost-minus-scorecard gap's spread (max minus min of factor-level means) was largest across drift (0.033), then shift (0.012), training size (0.003), and noise (0.002).

E. Internal validity and sensitivity

Across 20 repeated five-fold cross-validations, internal AUROCs were 0.906 (SD 0.003; scorecard), 0.894 (logistic regression), and 0.882–0.900 (default ensembles); nested-CV tuning gave 0.885, 0.903, 0.900. The deployed three-bin configuration was selected in 18 of 20 repeats; ensembles fitted in-sample at 0.991–1.000, falling 0.091–0.118 under CV. There was no

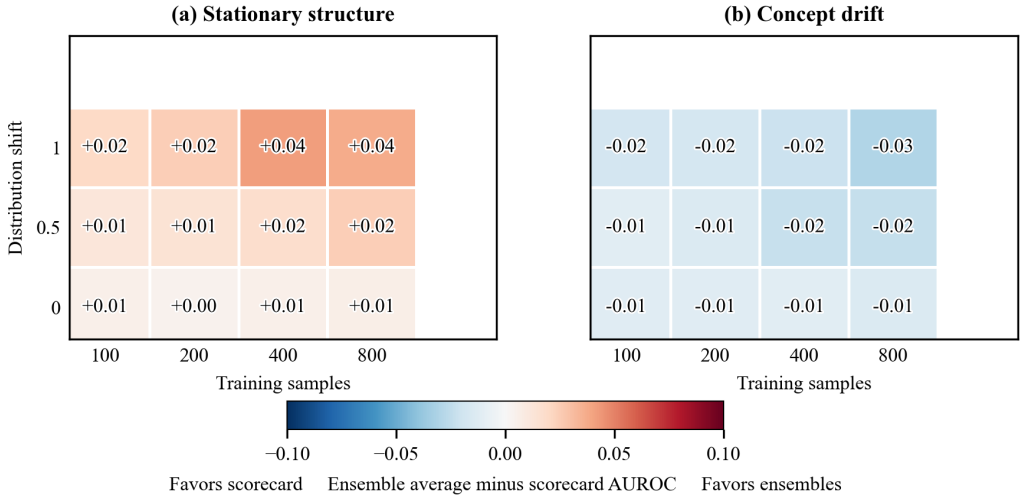


Fig. 3. Regime map: mean over 20 replicates of the ensemble-average minus scorecard test AUROC at 50% noise (12 of 24 cells per drift level); red favors the ensembles, blue the scorecard. Axes: $n \in \{100, 200, 400, 800\}$; shift $s \in \{0, 0.5, 1.0\}$.

internal advantage for the flexible models to lose: internal-minus-external deltas on the imputed West China cohort were −0.023 (scorecard), −0.042 (logistic regression), −0.005 (XGBoost), +0.008 (Random Forest), and +0.023 (CatBoost), and [3] reports the same null result across clinical prediction models generally.

IV. Discussion and Conclusion

Three results follow. First, complexity did not consistently improve external performance, but neither did the scorecard beat the simpler linear models. Seven integers matched the logistic regression and the formulas: the scorecard was numerically best on the complete cases (0.971, a 63%-prevalence estimate), not significantly differing from the logistic regression, ROMA, or XGBoost; the logistic regression led the imputed cohort and CPH-I the Japan test. The scorecard's honest comparator is the logistic regression, which ties or beats it on the imputed cohort, under MICE, under 30% noise, and in minimum harm at every weight; the scorecard's gain is hand-computability, at a small robustness cost. Second, on decisions: the published cutpoints missed 82 (ROMA, an imputation artifact) and 48 (CPH-I) of 188 imputed-cohort cancers versus 47 for the scorecard's tier; recalibrated, the scorecard stayed lowest among the three rules to 10:1, the logistic regression lowest overall to $w = 10$. Third, the simulation is supporting evidence only—one plausible mechanism (non-persisting learned interactions) by which flexible models lose their advantage under drift; the 22-of-24 result at $d = 1$ is close to built in (the drift-removed logit is additive in the features).

Limitations: (i) all three cohorts are Asian hospital series; (ii) HE4 was missing for 74% of West China patients and age for 85% of Japan patients; with HE4 fixed at its training median, ROMA's $N = 380$ sensitivity (56.4%) and AUROC (0.886) are imputation artifacts; the strongest result rests on 100 measured-HE4 patients; the Japanese controls are healthy volunteers rather than women with masses; (iii) West China's 49.5% prevalence far exceeds referral populations' 10–25% [9], tying Table II/Fig. 2 to it; (iv) no assay platform is named: the 13.1%/27.7% cutoffs are Fujirebio-specific (ARCHITECT: 7.4%/25.3%), as are the tertiles; (v) no ultrasound; (vi) the simulation is a simplification; here the scorecard matched the logistic regression and the formulas, beat two of the three tuned ensembles on the outcome-selected complete cases, hand-computability its asset.

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